

Script Type: DocType Event

Reference Clubbing Sheet

Document Type:

Event Frequency: All

DocType Event: Before Save

Script

```
# Server Script for Clubbing Sheet (runs on validate and on_submit)
# 'frappe' and 'doc' are available at the top-level scope only.
# IMPORTANT: Functions cannot call other functions in RestrictedPython — all logic self-contained.

def calculate_total_weight(doc):
    total = 0
    for item in doc.items:
        total += frappe.utils.flt(item.weight_kgs)
    doc.total_weight = total

def validate_and_set_load_type(doc):
    if not doc.items:
        doc.load_type = ""
        return

    customer_weights = {}
    for item in doc.items:
        if item.customer:
            customer_weights[item.customer] = customer_weights.get(item.customer, 0) + frappe.utils.flt(item.weight_kgs)

    customers = list(customer_weights.keys())

    # Rule: If any customer has >= 5000 kgs, it must be a Full Load for that customer only
    full_load_customers = [c for c, w in customer_weights.items() if w >= 5000]

    if full_load_customers:
        if len(customers) > 1:
            msg = frappe._("Customer %s has a total weight of %s kgs (>= 5000 kgs). Orders >= 5000 kgs must be clubbed separately as a Full Load.") % (full_load_customers[0], customer_weights[full_load_customers[0]])
            frappe.throw(msg)
            doc.load_type = "Full Load"
        elif len(customers) > 1:
            doc.load_type = "Part Load"
        elif len(customers) == 1:
            doc.load_type = "Full Load"
    else:
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doc.load_type = ""

def validate_route_compatibility(doc):
    if not doc.items or len(doc.items) < 2:
        return

    # -----
    # Route zones from Madurai:
    # N = North (NH44 towards Trichy, Salem, Chennai)
    # S = South (NH44 towards Virudhunagar, Tirunelveli, Kanyakumari)
    # W = West (Kerala route via Theni/Kumily)
    # E = East (towards Ramanathapuram, Rameswaram)
    # M = Madurai local — MUST be a separate load entirely
    # Conflict: N+S (opposite ends of NH44) or E+W (opposite east-west)
    # -----
    zone_map = {
        "M": ["Madurai", "Melur", "Usilampatti"],
        "N": [
            "Dindigul", "Tiruchirappalli", "Trichy", "Karur", "Namakkal", "Salem",
            "Erode", "Tirupur", "Coimbatore", "Pollachi", "Palani", "Oddanchatram",
            "Thanjavur", "Kumbakonam", "Nagapattinam", "Perambalur", "Ariyalur",
            "Mayiladuthurai", "Cuddalore", "Villupuram", "Pondicherry", "Puducherry",
            "Tiruvannamalai", "Dharmapuri", "Krishnagiri", "Hosur", "Vellore",
            "Kanchipuram", "Chengalpattu", "Chennai", "Nilgiris", "Ooty", "Gudalur",
            "Pudukkottai", "Tiruppathur"
        ],
        "S": [
            "Virudhunagar", "Aruppukottai", "Sivakasi", "Sattur", "Kovilpatti",
            "Tirunelveli", "Thoothukudi", "Tuticorin", "Nagercoil", "Kanyakumari",
            "Rajapalayam", "Tenkasi", "Thenkasi", "Courtallam", "Sankarankoil",
            "Srivilliputhur", "Padmanabhapuram"
        ],
        "W": [
            "Theni", "Periyakulam", "Cumbum", "Kumily", "Munnar", "Idukki",
            "Kottayam", "Alappuzha", "Alleppey", "Ernakulam", "Kochi", "Cochin",
            "Thrissur", "Palakkad", "Malappuram", "Kozhikode", "Calicut",
            "Wayanad", "Kannur", "Cannanore", "Kasaragod", "Mangaluru", "Mangalore",
            "Udupi", "Mysuru", "Mysore", "Bengaluru", "Bangalore", "Hassan",
            "Thiruvananthapuram", "Trivandrum", "Kollam", "Pathanamthitta",
            "Thodupuzha", "Muvattupuzha", "Changanacherry", "Pala", "Puttur",
            "Chikmagalur", "Kodagu", "Madikeri", "Virajpet", "Chamarajanagar",
            "Mandya", "Ramanagara", "Tumkur", "Kolar", "Shimoga", "Davangere",
            "Hubli", "Dharwad"
        ],
        "E": [
            "Sivaganga", "Ramanathapuram", "Paramakudi", "Manamadurai"
        ]
    }

```

```
}  
}  
  
def get_zone(city):  
    if not city:  
        return "?"  
    city_lower = city.lower()  
    for z in zone_map:  
        for c in zone_map[z]:  
            if c.lower() == city_lower or city_lower in c.lower() or c.lower() in city_lower:  
                return z  
    return "?"  
  
zones_present = []  
madurai_items = []  
for item in doc.items:  
    z = get_zone(item.party_location or "")  
    if z not in zones_present:  
        zones_present.append(z)  
    if z == "M":  
        madurai_items.append(item.party_location)  
  
# Rule 1: Madurai must be alone — cannot be clubbed with any other district  
if "M" in zones_present and len(zones_present) > 1:  
    frappe.throw(  
        "Madurai orders cannot be clubbed with other districts. "  
        "Please create a separate Clubbing Sheet for Madurai deliveries."  
    )  
  
# Rule 2: North + South conflict — vehicle must backtrack through Madurai  
if "N" in zones_present and "S" in zones_present:  
    frappe.throw(  
        "Route conflict detected! Orders going NORTH (e.g. Karur, Salem, Trichy) "  
        "and SOUTH (e.g. Virudhunagar, Tirunelveli) cannot be clubbed together. "  
        "The vehicle would have to backtrack through Madurai, wasting time and fuel. "  
        "Please create separate Clubbing Sheets for North-bound and South-bound orders."  
    )  
  
# Rule 3: East + West conflict — opposite directions  
if "E" in zones_present and "W" in zones_present:  
    frappe.throw(  
        "Route conflict detected! Orders going EAST (e.g. Ramanathapuram) "  
        "and WEST (e.g. Theni, Kerala) cannot be clubbed together. "  
        "Please create separate Clubbing Sheets."  
    )  
  
def validate_on_submit(doc):
```

SERVER SCRIPT

CLUBBING SHEET

```
if not doc.trip_id:
    frappe.throw("Please enter a Trip ID before submitting the Clubbing Sheet.")

def calculate_loading_sequence(doc):
    if not doc.items:
        return

    dm = {
        "Madurai": 0, "Melur": 30, "Usilampatti": 45, "Manamadurai": 60,
        "Sivaganga": 55, "Dindigul": 65, "Virudhunagar": 65, "Theni": 70,
        "Srivilliputhur": 75, "Aruppukottai": 80, "Sivakasi": 80,
        "Periyakulam": 85, "Tiruppathur": 90, "Oddanchatram": 95, "Sattur": 95,
        "Cumbum": 90, "Pudukkottai": 100, "Rajapalayam": 100,
        "Paramakudi": 130, "Kovilpatti": 130, "Palani": 120, "Ramanathapuram": 115,
        "Kodaikanal": 115, "Thenkasi": 115, "Tenkasi": 115, "Courtallam": 120,
        "Sankarankoil": 125, "Kumily": 130, "Gudalur": 130, "Munnar": 140,
        "Idukki": 145, "Karur": 140, "Tiruchirappalli": 135, "Trichy": 135,
        "Tirunelveli": 155, "Thoothukudi": 160, "Tuticorin": 160, "Perambalur": 160,
        "Ariyalur": 175, "Thanjavur": 175, "Pollachi": 180, "Kottayam": 185,
        "Namakkal": 200, "Alappuzha": 200, "Alleppey": 200, "Thodupuzha": 205,
        "Padmanabhapuram": 205, "Coimbatore": 210, "Tirupur": 215, "Kumbakonam": 215,
        "Ernakulam": 220, "Kochi": 220, "Cochin": 220, "Salem": 220,
        "Nagercoil": 230, "Erode": 240, "Muvattupuzha": 235, "Muvuttupuzha": 235,
        "Pala": 195, "Changanacherry": 195, "Pathanamthitta": 215,
        "Kollam": 250, "Quilon": 250, "Palakkad": 250, "Mayiladuthurai": 250,
        "Kanyakumari": 250, "Nilgiris": 265, "Ooty": 265, "Nagapattinam": 280,
        "Thrissur": 280, "Thiruvananthapuram": 290, "Trivandrum": 290,
        "Malappuram": 320, "Dharmapuri": 320, "Wayanad": 345, "Cuddalore": 365,
        "Kozhikode": 360, "Calicut": 360, "Villupuram": 370, "Mysuru": 370,
        "Mysore": 370, "Chamarajanagar": 390, "Mandya": 390, "Hosur": 385,
        "Pondicherry": 395, "Puducherry": 395, "Hassan": 395,
        "Tiruvannamalai": 400, "Virajpet": 405, "Kanchipuram": 440, "Vellore": 410,
        "Chengalpattu": 420, "Ramanagara": 420, "Kodagu": 420, "Madikeri": 420,
        "Kannur": 430, "Cannanore": 430, "Kolar": 430, "Bengaluru": 445,
        "Bangalore": 445, "Chikmagalur": 450, "Chennai": 455, "Tumkur": 475,
        "Puttur": 480, "Kasaragod": 490, "Mangaluru": 490, "Mangalore": 490,
        "Shimoga": 485, "Davangere": 510, "Nellore": 520, "Udupi": 530,
        "Tirupati": 530, "Hubli": 600, "Dharwad": 610, "Guntur": 650,
        "Vijayawada": 680, "Hyderabad": 770, "Warangal": 850, "Vizag": 970,
        "Berhampur": 1520, "Brahmapur": 1520, "Bhubaneswar": 1650,
        "Cuttack": 1680, "Puri": 1700, "Sambalpur": 1800, "Rourkela": 1850,
        "Krishnagiri": 355, "Pune": 1250, "Mumbai": 1450
    }

    for item in doc.items:
        city = item.party_location or ""
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dist = 0
if city in dm:
    dist = dm[city]
else:
    city_lower = city.lower()
    for key in dm:
        if key.lower() == city_lower:
            dist = dm[key]
            break
    if dist == 0:
        for key in dm:
            if city_lower in key.lower() or key.lower() in city_lower:
                dist = dm[key]
                break
    item.distance_from_madurai = dist

sorted_items = sorted(doc.items, key=lambda x: frappe.utils.flt(x.distance_from_madurai), reverse=True)
n = len(sorted_items)

if n == 1:
    sorted_items[0].loading_sequence = "Inside"
elif n == 2:
    sorted_items[0].loading_sequence = "Inside"
    sorted_items[1].loading_sequence = "Outside"
else:
    sorted_items[0].loading_sequence = "Inside"
    sorted_items[n - 1].loading_sequence = "Outside"
    center_num = 1
    for i in range(1, n - 1):
        sorted_items[i].loading_sequence = "Center " + str(center_num)
        center_num = center_num + 1

for i in range(len(sorted_items)):
    sorted_items[i].idx = i + 1
doc.items = sorted_items

# --- Run on validate ---
calculate_total_weight(doc)
validate_and_set_load_type(doc)
validate_route_compatibility(doc)
calculate_loading_sequence(doc)

# --- Additional check on submit ---
if doc.docstatus == 1:
    validate_on_submit(doc)
```

Request Limit:

5

Time Window (Seconds):

86400